MZI (integrated optics project)

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Chapter 1

Introduction

1.1 Material Selection

The material parameters used in the simulation are summarized in Table 1.1.

Table 1.1: Material and optical parameters

Parameter	**Material**	**Refractive Index**
n_1 (core)	Lithium niobate ($LiNbO_3$)	2.138
n_2 (cladding)	Silicon dioxide (SiO_2)	1.444
Operating wavelength	1550 nm	—

1.2 Checking single-mode condition

If waveguide width

$$w = 0.45\,\mu\text{m}$$

For an ideal symmetric slab waveguide, define

$$V = \frac{2\pi}{\lambda} w \sqrt{n_1^2 - n_2^2}.$$

For this definition, the first higher-order mode reaches cutoff at $V = \pi$. Therefore, the ideal single-mode criterion is:

$$\boxed{V < \pi}$$

Substituting:

$$V = \frac{2\pi}{1.55}(0.45)\sqrt{2.138^2 - 1.444^2}$$

$$\boxed{V \approx 2.876 < 3.142}$$

selected values satisfy this criterion, although they are relatively close to the first higher-order mode cutoff.